A2: Assignment Midterm Game

PixeledThoughts

GBDA 302: Global Digital Project 2

Group 8A- Section 1

Group Members:

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Process Overview Visualization (Group Work Only)

Please refer to the bottom of the doc for the process overview visualization.

Role-Based Process Evidence

Refer to screenshots below for evidence.

Vrunda Popawala - Researcher, Coder & Editor

Vrajani Popawala - Researcher, Coder & Editor

Ayushi Lad - Researcher, Coder & Editor

Anushka Kshirsagar - Researcher, Idea generation, Promoter, editor

Anisha Thiruselvam - Researcher, Coder & Editor, & Image Prompter

Sukena Abbas - Researcher, Editor, & Idea Generation

Goal of Work Session

For A2, our goal was to make a game that represents schizophrenia in a way that actually makes people feel something, not just learn a fact about it. We wanted players to experience what disorganized thinking can be like, even just for a few minutes. So we built a pixel-art game where you walk around a small town, go into stores, and try to put together a sentence while words are flying around, intrusive thoughts keep popping up, and a timer is stressing you out. All of that was intentional as it's meant to mirror the kind of cognitive overwhelm that can come with schizophrenia. On top of the concept, we also wanted the game to actually look good and feel cohesive, so a big part of our work session was making sure the pixel aesthetic, the characters, and the world all came together in a way we were proud of.

GenAI Documentation

GenAI was used for this task.

Entry Header: Vrunda Popawala

Roles: Coder & Editor

Primary responsibility for this work: Adding features (i.e. return home button), images, making the player move, improving the code when it breaks, connecting the code.

Goal of Work Session: The goal was to use ChatGPT to code our game. Over a few days I focused on asking for detailed prompts, iterating the code and asking for help when I ran into any problems.

Date Used: March 2nd-5th, 12th 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: I asked specific prompts to help code my game, fix code when it breaks or ask for help.

Summary of Interaction: I gave ChatGPT specific prompts on what I am looking for, for example "I want to add a background that will replace the building, path nd the grass". Based on what I gave, I either take the code or keep iterating until I'm satisfied with what I'm looking for.

Human Decision Point(s): After generating the initial code, I ran it, whether the output matched my intended design and visual direction. At this stage, I made several key decisions based on what I observed. For example, I adjusted elements such as colour choices, x and y coordinates, and sizing to better align with my design goals and improve visual balance.

I also assessed whether the code clearly communicated my idea and met the assignment requirements. If something felt off such as poor image spacing, unclear hierarchy of the text , or lack of visual appeal I iteratively modified the code and re-ran it.

In addition, I made design decisions, such as selecting colours that felt more cohesive and adjusting positioning to create a more polished and intentional layout. This process involved trial and error, where I continuously compared outputs and refined the code until I achieved a result that was both functional and visually effective.

Integrity & Verification Note: I ran all the code it gave, ran it by my team members to make it what we were looking for. Keep iterating and modifying the code.

Scope of GenAI Use: To code the game, fix any broken codes and ask for help when I run into a problem.

Limitations or Misfires: When asking to fix a broken code, it tends to get confused on what I'm asking to fix therefore I had to keep iterating multiple times or try to fix the code myself.

Summary of Process (Human + Tool)

- Put the code into chatGPT, to help with errors.
- Debug the code when it breaks.
- Ask prompts to create the code or for any clarification.
- Iterating my prompts to get what I'm exactly looking for.

Decision Points & Trade-offs

- Changing the colour myself, the sizing of the buttons, and fixing the x, y coordinates.
- Taking suggestions or not.

Verification & Judgement

- Running the code each time I made a change.
- Asking my team member to take a look at a change I made.
- Made sure it aligns with our game expectations.

Limitations, Dead Ends, or Open Questions: Since no one in our group are coders we struggled to fix problems when we ran into them so I kept having to ask multiple times which was frustrating at times. Or when I run into problems I ask to explain it, it won't always do a good job explaining it.

GenAI link to Transcript

https://chatgpt.com/share/69a9ec8b-9cf4-800e-91a9-014570f9ad36

https://chatgpt.com/share/69a77fd2-1b98-800e-b849-128a7cc6c79e

https://chatgpt.com/share/69b2e164-d454-800e-a212-6cb71514d857

***Entry Header*:** Vrajani Popawala

Roles: Coder and Editor

Primary responsibility for this work: The responsibility for this game was coding the game mechanics, the levels, games timer, buttons.

Goal of Work Session: To complete a game that would teach schizophrenia. Complete 60-70% of the game before the playtesting.

Date Used: March 2, 3, 4, 5, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: The purpose was to help me create the code based on our prompts to create the game mechanics. I reiterated the prompts over a few days to improve the code and clarified what problems I run into and explained what the codes meant.

Summary of Interaction: I gave GenAI specific prompts on what I wanted such as "create button on the game screen so the player can leave the screen". Then GenAI gave the code for

the prompt. I would give the GenAI the specific code I already had because sometimes the given code wouldn't match what was in the original code I had. I would also give the code to GenAI and then ask from there where I would add the new function, why it is not working, and give the full code with the new function added.

Human Decision Point(s): I would check all the code given and make sure it follows the prompt I gave and visually aligns with what I wanted. It would give suggestions on the game functions and I would accept or reject the ideas. I had to iterate GenAI prompts a lot as the code given wouldn't always work. I would ask GenAI what code of line is doing a specific function and then change it manually which saves a lot of time rather than putting the whole code into GenAI again. That specifically helped with positioning specific elements. For example when creating the button it wasn't in the right spot so I GenAI what code was in charge of it and then change it myself.

Integrity & Verification Note: I tested all the code given by GenAI. All the prompts were made by me and reviewed by me. The game was tested to make sure it would run smoothly and didn't break the code. I checked with my group's members to make sure we all agreed that this was wanted and reviewed the research.

Scope of GenAI Use: GenAI was used to code the game. GenAI fixed the code, gave the functions and explained the code when needed.

Limitations or Misfires: GenAI would break the code when asked to make a new game feature. I would have iterated a lot to make the game work. The GenAI would get confused on what I wanted due asking too much in one chat box, so that is why I would in a new chat to fix a code of line to reduce confusion.

Summary of Process (Human + Tool)

- Gave GenAI my prompts
- Asked GenAI to explain the code when it wasn't working
- Reviewed the code and made sure it aligned with what I wanted

Decision Points & Trade-offs:

- Fixed the colours and position myself
- Made sure key features like the buttons and timer would work the play testing game

Verification & Judgement

- Verified all the code given by GenAI by playing the code and making sure it didn't break the code or other parts of the game
- Made sure interactive parts like buttons would work

Limitations, Dead Ends, or Open Questions: GenAI would give codes that would break the game. Even when I paste the whole code into the GenAI and ask why it wasn't working, it would still break the game.

GenAI link to Transcript

https://chatgpt.com/share/69b1f135-eaa0-800f-b01d-98dd57bfd614
https://chatgpt.com/share/69b1f167-0c30-800f-95e5-d2e379b2ca3f
https://chatgpt.com/share/69b1f1a1-d858-800f-9b58-986985e8010f
https://chatgpt.com/share/69b1f1fb-b1b0-800f-bfc8-6da6886b08d4
https://chatgpt.com/share/69b1f216-3898-800f-8385-2b35c2fea090

***Entry Header*:** Ayushi Lad

Roles: Coder and Editor

Primary responsibility for this work: Game mechanics and level mechanics, → locks, levels, speech bubble, try again/fail screen, instructions, level backgrounds, stars

Goal of Work Session: Complete 70/80% of the game

Date Used: March 2, 3, 4, 5, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: The purpose of using ChatGPT was to code different aspects of our game and level mechanics. Over the past few days, I focused on prompting ChatGPT with a variety of these, including the instructions pop-up and the stars that indicated when a level was complete.

Summary of Interaction: I prompted ChatGPT with different aspects that I wanted to include in our game to make it work smoothly and efficiently. I iterated with ChatGPT because its outputs sometimes didn't work. I would copy and paste the full code of our existing game, including the various .js files and the .index files and paste it into ChatGPT with a prompt of the element I want. For example, I would say "I have a lock image that i want to place over each of the sublevels 2 and 3 for each building. once the player comeplete level 1 sublevel, the next time they click that same building level two will be unlock with the lock image no longer showing.", and attach the full code. Moreover, I would also copy and paste the code that was broken when I would add elements and ask for fixes.

Human Decision Point(s): A lot of the code ChatGPT gave me needed to be edited, especially when it came to more UI focused aspects of the game rather than mechanics. For example, when generating the stars to indicate when a level was complete, the coordinates that ChatGPT gave me did not align with the location of the various buildings. In this situation, I had to manually change them in the code. Moreover, when working with the locks on the levels screen, the locks initially showed up in the corner of each level block rather than in the center. When trying to figure out how to center the image with ChatGPT, it did not work that well. So, I went into the code and manually played with the numbers in the code that altered the image's x and y coordinates as well as its size in order to get the desired outcome. Other than positioning, I also made the decision to edit code for stylistic purposes. When ChatGPT gave me game elements that were more UI focused, such as the instructions

pop-up, I would go in and edit the function code to change the default instructions it had generated, as well as the colours it gave.

Integrity & Verification Note: I would copy and paste the code given by ChatGPT and test it out in VSCode, and go back and forth using that process until I got the desired outcome.

Scope of GenAI Use: I used ChatGPT to code the game.

Limitations or Misfires: Sometimes ChatGPT misunderstood what I was asking and gave code that didn't do what I wanted. It would also give code that would sometimes break other aspects of the game.

Summary of Process (Human + Tool): I put in the whole code of our existing game and put it into ChatGPT, along with a prompt of a game mechanic or element that I wanted to implement. For example, for the locks on the game, I put in the full code and prompted ChatGPT to provide me code that would add an image of the lock on top of levels 2 and 3 and make it so the user cannot click them, and the levels would only unlock after level 1 was complete. I would then check the code in VSCode and see if it worked. At first, it didn't work, so I had to go back and forth until it did, and then I went in and edited the code to change the sizing and position of the lock image. I kind of used the same kind of process for each thing I added.

Decision Points & Trade-offs: I mainly made stylistic changes like colours, wording and positioning of certain elements of the game.

Verification & Judgement: Every time I asked ChatGPT to give me a new element or game mechanic to add to our existing code, I tested it out in VSCode to verify that it did what I wanted.

Limitations, Dead Ends, or Open Questions: Sometimes ChatGPT gave code that wouldn't work or would work but break other aspects of the game, so I would have to go back and ask ChatGPT to find a way to make things work.

GenAI link to Transcript

https://chatgpt.com/share/69a77ff7-5a74-8000-81b1-8faf8598da0a
https://chatgpt.com/share/69b1f856-41bc-8000-abbe-6973320bbeda
https://chatgpt.com/share/69b1f15e-8238-8000-a52b-359da83b58be

***Entry Header*:** Anisha Thiruselvam

Roles: Coder, Editor & Image Prompter

Primary responsibility for this work: Generate visuals, upload them to the game, fix the screen resizing and fix game visuals with coordinates.

Goal of Work Session: To complete 70% of the game and to figure out the main game mechanics.

Date Used: March 3, 4, 5

Tool Disclosure: ChatGPT 5.2, Adobe Firefly Gemini 3.1 (w/ Nano Banana Pro)

Purpose of Use: ChatGPT 5.2 was used to understand and debug the written code. It was used to generate code to create the desired effect. Firefly was used to create the visuals for the game.

Summary of Interaction: I input the full code that needed changes. I then asked ChatGPT to modify the code to incorporate the images based on the file names and make the screen responsive regardless of what laptop is being used. Some code was added, but the majority of it led to bugs, so it had to be modified manually. There were also smaller tasks I had asked it to do, such as create boundaries so the characters do not walk over them. For the image generation, I would give a clear prompt and generate as much as needed to get the desired look. Once that was achieved, I imputed that image to generate more images in the same style.

Human Decision Point(s): A lot of the provided code did not run immediately. It required a lot of back and forth, and required me to take only certain parts of the code and discard the rest. Sometimes the given code gave me different ideas on how to make the game function the way I want in an easier way, such as moving the screen positioning over instead and having set values, so that the screen will be responsive. When it came to resizing the screen positioning, I had to use my prior knowledge in code and added variables to fix the positioning rather than pixels. For the visuals, I had to regenerate images if they did not fit the theme we wanted to go with. I figured out how to make efficient prompts to get the desired image. Sometimes I had to bring the images into Figma and manually make design choices and changes to make the image look better.

Integrity & Verification Note: The code was run to ensure that it worked and did not crash or give an error. The images generated were checked for spelling and if they visually made sense (ie, correct number of arms, etc.).

Scope of GenAI Use: GenAI was used to enhance the game's overall visuals and game mechanics.

Limitations or Misfires: ChatGPT often gave code that caused errors, or Firefly generated images that were not what was desired. I had to disregard some of it, especially a lot of the suggested code for the background resizing.

Summary of Process (Human + Tool): I gave ChatGPT the code that needed to be fixed, and then when it gave out code, I manually put in the code to see if it ran. Oftentimes, there were errors that I needed to change or fix. I would also have an idea of what type of code I wanted to use, and would explain that to ChatGPT, and it would then give me a working code to implement, based on the logic I suggested. I would try to impute smaller sections to see if it worked and work accordingly. For Firefly, I prompted Gemini to create me images with a theme. I would give a more specific prompt if needed, depending on whether it was not accurate enough. I then would put in the generated picture I was happy with, and code other

elements in the same outline. To put a lot of the graphics on the screen together, I would use Figma to edit them, showing humans designing.

Decision Points & Trade-offs: I had to either choose to fix what ChatGPT offered or ask it to generate a different solution for it. I had to trade off the collisions of the building and the person walking because I was not able to fully incorporate that in without the game crashing.

Verification & Judgement: Ran the code to ensure everything worked and tried playing the game to make sure the next screens of the game did not get affected. I made sure the images looked the way I wanted.

Limitations, Dead Ends, or Open Questions: Was not able to set the boundaries for the buildings. Could not figure out how to properly make the game the entire screen of a game, without it being stretched based on different laptops. There were a lot of adjustments that needed to be made to incorporate the code or images. The images would not have the proper text I asked for at times.

*Refer to the end of the appendix for the screenshots of the Firefly images
GenAI link to Transcript

https://chatgpt.com/share/69a78413-3bb0-8004-b55b-460ea8dcb155
https://chatgpt.com/share/69a8d069-b00c-8004-aeca-bdd1c9e3822b

***Entry Header*:** Anushka Kshirsagar

Roles: Idea generation, Promoter, editor

Primary responsibility for this work: All ideas were done by us, the pixelized aesthetic, the 3 characters. I used ChatGPT to help me write code for all my ideas, such as the pixel background, 4 housing placement, and character moving around the world, and promoted it such that it gives me exact code that matches my game design aesthetic

Goal of Work Session: The goal of this work session was to begin building the playable version of the game environment for A2 midterm. I wanted to translate the visual and interaction ideas into working code so the player could move through a pixel world and interact with different locations.

Date Used: March 3rd, March 4th, March 5th, March 10th

Tool Disclosure: ChatGPT 5.2

Purpose of Use: I used ChatGPT to help generate JavaScript and p5.js code that could implement my design ideas. The tool helped with structuring all the code for the world map, character movement, background elements, and basic gameplay mechanics. Code also helped me properly prompt the sentences better for making sure its in “ai language” for better results

Summary of Interaction: Gave ChatGPT detailed promotion and I described the type of game I wanted to create, including the pixel art aesthetic, character movement, and building layout. I also provided references and detailed prompts explaining the design style. ChatGPT generated all code suggestions, which I tested and modified in my project. In some cases the initial code did not work correctly, so I refined my prompts and adjusted the code until the result matched my actual design value. In some cases, the initial code did not work correctly or didn't fully match what I had in mind, so I refined my prompts, clarified my requirements, and was very very unique and nitpick about design requirements and code requirements. Many time I had to break the problem into smaller parts and ask for more specific help (like movement logic or collision handling).

Human Decision Point(s): All decisions were made by me, but I actively chose which parts of the generated code to keep, modify, edit, or completely discard depending on how well the output aligned with my design goals. When the generated code did not match my design needs or didn't align with my other group members' expectations or their code output, I decided not to use that code. Instead, I compared it with my teammates implementations, sometimes copy-pasting their code to better understand their approach, and then asked ChatGPT to generate a similar algorithm based on that structure. I also made adjustments to consistency across our project, such as matching variable names, logic flow, and formatting style. In some cases, I combined ideas from both the generated code and my teammates code to create a more refined solution. This process helped me stay in control of the final outcome rather than relying entirely on generated results.

Integrity & Verification Note: All generated code was reviewed and tested by my group members and me before being included in the project. I verified that code worked correctly and made sense in terms of game mechanics, the core creative and design thinking was done by our group… I also made sure that chatgpt prompts are valid and make sense for human (us) and ai language

Scope of GenAI Use: ChatGPT was used as a coding tool to help the game design come to life. It was not used to generate the original concept, gameplay idea, or design decisions

Limitations or Misfires: Many code lines did not run properly, and at first, it gave blank white screens or just didn't run in general, also gave a lot of red lines (code running errors) in VS Code, which created a lot of debugging and manual iterations

Summary of Process (Human + Tool): I first planned the game design, then made a detailed prompting sentence, I asked ChatGPT to add onto the prompt and make it more detailed, then copy pasted the prompt again to give a detailed code.

Decision Points & Trade-offs: One key decision was promoting myself and then asking ChatGPT to add onto that prompt. I made sure that my original idea stayed the same but I denied the ideas that were given by ChatGPT.

Verification & Judgement: To make sure the player could move, the environment displayed correctly, and the gameplay interactions worked exactly as planned, I tested the game in the browser. Only the code that fit my idea and executed well stayed.

Limitations, Dead Ends, or Open Questions: Dead ends were when there was code given that looked correct, but it didn't run in VS Code. We had to manually fix the code because even though ChatGPT said it was right, when we copy/pasted it, it just didn't work. This was frustrating because sometimes the errors weren't obvious, and we had to spend extra time figuring out small issues like syntax mistakes, missing parts, or things that didn't match our setup. Another problem was that some of the code was too general or didn't fully match our project, so I had to keep rewriting my prompts to be more clear and specific. There were also times where fixing one problem caused another one, so it felt like a lot of trial and error. Because of this, I couldn't fully rely on the generated code and had to use my own understanding and also check with my group members to make sure everything actually worked.

GenAI link to Transcript

https://chatgpt.com/share/69a6028c-1bc8-800d-8272-b7479885cad2

https://chatgpt.com/share/69b2d4be-d9f4-800d-b76e-6aade4a610e5

https://chatgpt.com/share/69b2d4d0-a63c-800d-a1d6-6acd20db6972

***Entry Header*:** Sukena Abbas

Roles: Researcher, Editor, & Idea generation

Primary responsibility for this work: My main responsibility was to help generate ideas for the game concept, research schizophrenia experiences, and assist with early coding development.

Goal of Work Session: The goal of these work sessions was to first develop an idea for the game and later start building early versions of the game mechanics and structure.

Date Used: February 28, March 2, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: I used ChatGPT to help generate ideas for the game concept and to explore possible mechanics related to schizophrenia experiences. Later, I used it to help create early coding examples, ask questions when the code was not working, and help organize the structure of the game.

Summary of Interaction: I asked ChatGPT for help brainstorming ideas for games that could represent schizophrenia experiences. One idea I explored was the Soundscape Maze Game (Echoes in the Maze). In this concept, players move through a maze by listening to sound cues. Some sounds guide the player correctly while others mislead them, representing auditory hallucinations. Players have to decide which sounds to trust while managing their stress level, and too much stress would make the sounds more distorted and confusing. After brainstorming these ideas, the team discussed them and decided to move forward with a

different concept that focused more on intrusive thoughts and sentence-building in social situations. Later in the project, I used ChatGPT again to help generate early versions of code for the game. I asked for help creating simple game structures, multi-file organization, and basic mechanics so we could start testing gameplay ideas, such as scrambled word selection, disappearing words, intrusive thought clicks, timer management, and level progression. ChatGPT also assisted in improving visual alignment, backgrounds, and adding a player character for better game presentation.

Human Decision Point(s): I went through all the code from ChatGPT and made sure it matched the ideas I had for the game. I decided which suggestions to keep and which to change to fit the look and feel I wanted. When I didn't understand what a line of code was doing, I asked ChatGPT to explain it and then made the adjustments myself, like changing colours, moving things around, or fixing text alignment. I also made decisions about how closely the generated ideas actually represented the experience of schizophrenia. For example, I had to decide whether certain mechanics, like the intrusive thoughts or the moving words, actually represented the experience we were trying to show or if they made the game too confusing or frustrating for the player. I chose to simplify or remove features that made the game harder to understand, even if they were interesting. When ChatGPT suggested multiple ways to implement a feature, I chose to select the option that would fit best with the existing code, rather than choosing the most difficult solution. I also had to decide when to stop iterating with GenAI and instead manually fix the code myself, especially where small changes were needed that I could do.

Integrity & Verification Note: I tested all the code from ChatGPT to make sure the game ran properly and didn't crash. I wrote all the prompts and carefully reviewed the suggestions before using them. I also checked in with my group members to make sure we all agreed on the game design and that our research was accurate.

Scope of GenAI Use: I used ChatGPT to help write and fix the code for the game. It suggested functions, explained how lines of code worked, and gave ideas for mechanics. I made the final decisions and implemented the changes myself.

Summary of Process (Human + Tool): I started by using ChatGPT to brainstorm ideas for the game and explore different ways to represent schizophrenia through gameplay. After choosing a direction with my group, I used ChatGPT to generate basic code structures and mechanics, such as word selection, timers, and interactive elements. I would then take that code and test it in VS Code to see if it worked. If it didn't, I went back to ChatGPT to debug or explain the issue, and repeated this process multiple times. Along the way, I made manual adjustments to improve visuals, positioning, and overall gameplay. Throughout the process, I combined ChatGPT's suggestions with my own decisions to make sure the game stayed aligned with our concept and worked properly.

Decision Points & Trade-offs: I had to decide which game features were important and which were optional for the first version. I chose simpler mechanics for the early prototype to make testing easier. I also balanced how chaotic the game felt with how playable it was, because too much chaos could confuse players.

Verification & Judgement: I tested every part of the game, like the word selection, timer, and intrusive thought interactions, to make sure they worked correctly. I decided which suggestions from ChatGPT improved the game and which ones needed tweaking. I also made sure the game reflected the experiences we wanted to show while still being playable and understandable.

Limitations, Dead Ends, or Open Questions: Sometimes the code would break when ChatGPT tried to add new features. I had to make multiple adjustments to get it working properly. ChatGPT could also get confused if I asked too many things at once, so I would focus on one issue at a time or start a new chat. Some suggestions still need more testing to make sure they work well in the final version of the game.

GenAI link to Transcript:
https://chatgpt.com/share/69b2e253-7e2c-8008-b941-e497857a81b4

References

[1] Adobe. 2026. Illustration of 4 buildings in a street in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[2] Adobe. 2026. Illustration of a boy character in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[3] Adobe. 2026. Illustration of a cafe in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[4] Adobe. 2026. Illustration of a choose your characters screen in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[5] Adobe. 2026. Illustration of a flower store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[6] Adobe. 2026. Illustration of a girl character in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[7] Adobe. 2026. Illustration of a hat, teddy bear and a shirt in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[8] Adobe. 2026. Illustration of a levels screen in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[9] Adobe. 2026. Illustration of a lock in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[10] Adobe. 2026. Illustration of a non-binary character in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[11] Adobe. 2026. Illustration of a rose flower, a sunflower and a bouquet of assorted flowers in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[12] Adobe. 2026. Illustration of a strawberry shortcake, a coffee and a croissant in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[13] Adobe. 2026. Illustration of a “You have failed” screen in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[14] Adobe. 2026. Illustration of clothing in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[15] Adobe. 2026. Illustration of the “Pixelated Thoughts” home screen in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[16] Adobe. 2026. Illustration of the interior of a book store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[17] Adobe. 2026. Illustration of the interior of a clothing store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[18] Adobe. 2026. Illustration of the interior of a coffee store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[19] Adobe. 2026. Illustration of the interior of a coffee store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

[20] Adobe. 2026. Illustration of the interior of a flower store in pixel art. Generated with Adobe Firefly (Gemini 3 w/ Nano Banana Pro). https://firefly.adobe.com/generate/image

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Appendix

Feedback we got back during the in-class showcase.

Noticed while gameplay

- Confused on how to play with the thoughts game, instructions near here
- Failed a lot on the third level
- Back button to get out of this
- Boundaries for intrusive thoughts should be fixed
- There's a missing "A" third level in the pixel books
- Gave 3 stars even when she didnt finish the level?

feedback

- A button where on the top where you have the instructions that can be clicked on
- Make the flowers related to the sentences
- Speed is good
- Try again screen – include a return home
- Feels like mini games
- There should be an overarching goal. Maybe like out on a shopping day and try to decide stuff. And then at the end it can show what you bought.
- A bigger storyline – better objective
- Make it so the player can click on the character
- A lot of people think its dyslexia
- Get a win screen after winning

During gameplay and feedback:

- The player initially thought they had to type the sentence, and did not realize they needed to click words to form the sentence.
- Some players were unsure what the goal of the game was, and it felt like a series of small mini-games rather than one clear objective.
- A suggestion was to include a help or instruction button during levels, in case players forget how to play.
- The player completed the first two levels quickly.
  They noticed that the timer gets shorter with each level, which increases the difficulty.
- By the third level, the player struggled more and had to retry several times.
- The flower level felt confusing, mainly because of the word choices.
- Some levels felt tricky due to the number of words and distractions.
- Players felt that the words on screen were overwhelming, especially when moving and combined with intrusive thoughts.
- One intrusive thought message ("Did you forget…") appeared partially off-screen.
- A player tried pressing certain buttons but nothing happened, which caused some confusion.
- A suggestion was to add clearer icons or visuals for levels, showing what the player is trying to ask for or buy.

Research Questions that were made in the group charter

Vrajani

Research question: How Do Auditory and Visual Hallucinations Actually Present in Real Experiences?

Auditory and visual hallucinations are sensory experiences that feel real to the person experiencing them. They are common features of schizophrenia, and understanding how they feel helps explain the challenges many individuals face [23][29]. Auditory hallucination involves hearing sounds, voices, or conversations that others do not hear. These experiences include hearing one or more voices talking, voices that may seem to come from outside one's head, and the experience feels like a real sound, not just a vague thought. In schizophrenia, auditory hallucinations are the most common. These voices can be distracting and can interfere with a person's attention and ability to focus in the real-world [23]. Visual hallucinations involve seeing things that are not actually there. It is more than just seeing things, hallucinations can involve any one or all of your senses: sight, sound, smell, touch, and taste. These visual experiences can range from simple shapes or flashes of light to complex images such as people or light [29].

Vrunda

Research question: What Are the Core Symptom Categories of Schizophrenia — and How Do They Differ?

Schizophrenia is a chronic brain disorder that affects how a person thinks, feels and behaves, and when it is active it includes symptoms such as delusions, hallucinations, disorganized speech and thinking and lack of motivation. The experience of schizophrenia often includes symptoms that disrupt a person's sense of reality and information processing, including false beliefs and sensory perceptions that others do not have. These symptoms can make it difficult to distinguish what is real and can interfere with everyday functioning such as work, relationships and self care. While there is no cure, treatment can significantly reduce symptoms and help people manage the condition. Understanding these core symptom categories helps explain why schizophrenia is complex and why people with it may struggle with daily social challenges [22] [26].

Ayushi
Research question: How Does Schizophrenia Affect Thought Organization and Decision-Making?

Organization
- They have disorganized thinking, which is observed as jumbled or irrelevant speech, making it difficult for them to talk to people [31]
- For example, they may answer something different from what they are asked [26]
- They have disorganized behaviour, making it difficult to do a task; a person does things that appear bizarre or purposeless, behaviour could be unpredictable or inappropriate [31]
    - Laughing excessively, staying very still [24]

Decision Making
- Delusions make it difficult to decide, as individuals may believe things that aren't true, even with clear evidence [31]
- Decisions could be affected by hallucinations as the person may hear, smell, see, touch or feel things that aren't there [31]

Sukena
Research question: What is the Emotional and Social Experience of Living With Schizophrenia?

People with schizophrenia often face strong emotional challenges internally, but have trouble expressing them externally, which can make it hard for others to understand how they feel. This can create stress for them and make daily life a lot more challenging, as they may have difficulty recognizing and regulating their emotions. Socially, schizophrenia can make social interaction difficult, leading to struggles to make and keep friends or interact confidently with others because of stigma and difficulties understanding social cues. Poor social functioning and lower quality of life are common; however, with strong support from family, friends and community programs, one can improve social connection and well-being. [30][31]

Anisha
Research question: How is Schizophrenia Commonly Misrepresented in Media and Games?

There is an issue with depictions of schizophrenia because they are often stereotyped or characterized by misinformation about what the treatments, causes, or symptoms are. This is due to the lack of study done when it is portrayed in films. In a study, two researchers rated the accuracy of the characters with schizophrenia, where it was concluded that the cinematic depictions were not supported by research. The symptoms shown often in movies were of caucasian men, with unpredictable behaviour, dangerous violence, delusions and a lot of self-harm/suicide. Some of the misinformation was that those with schizophrenia are homicidal maniacs, and creates a us vs them mentality, making it seem as though they should be feared or avoided, which can be very damaging to their community. Other inaccuracies

include that this disability has a higher rate in African Americans compared to Caucasians, and that their suicidal rate is fairly low [28].

Anushka

Research question: What Game Mechanics Can Simulate Cognitive or Perceptual Distortion Without Being Exploitative?

Game mechanics can model cognitive or perceptual distortion in a non-exploitative way by making the distortion the explicit learning target rather than a tool to force engagement or spending. For example, the serious game Silver has players classify characters' thoughts as helpful or distorted, which improved adolescents' recognition of cognitive distortions and reduced their own biased thinking (de Korte et al., 2024). Designers can also safely simulate unstable perception using temporary visual or auditory misperceptions that clearly reset and are framed as part of the narrative challenge, drawing on work about Game Transfer Phenomena that maps how game-related perceptual distortions can be experienced without necessarily causing distress (Ortiz de Gortari & Griffiths, 2022). To avoid exploitation, these mechanics should avoid dark patterns like deceptive rewards or artificial time pressure and instead emphasize transparency and player autonomy in how distortions are introduced and resolved [25][27].